

Spare Parts Management Module Specification

Overview

This module manages automotive spare parts across procurement, storage, usage, and valuation. It supports day-to-day garage operations with accurate stock, fast lookup, and seamless links to jobs and billing.

Purpose

- Maintain real-time inventory by location.
- Ensure parts availability via reordering.
- Provide traceability for issues, returns, and warranties.

Core Features

- Inventory tracking: multi-location, bin-level, batch/lot, serials where applicable.
- Stock levels: min/max, safety stock, reorder points, lead times.
- Automated reordering: PO suggestions, supplier selection, approval workflow.
- Supplier management: vendor records, price lists, lead times, terms.
- Parts catalog: categories, OEM/aftermarket, make/model/year fitment, attributes, images, barcodes.
- Search & lookup: keyword, SKU/OEM, category, fitment, barcode scan, substitutes/alternatives.
- Valuation: FIFO/Weighted Average; per-location costing.
- Audit trail: movements, user, timestamp, reason codes.

Key Workflows

- Receiving parts:
 - Match PO, record quantities, lot/serial, cost, taxes, update stock, print labels.
- Issuing parts to jobs:
 - Reserve/issue to work orders, track used vs returned, push to invoice.
- Returns and warranties:
 - Customer returns (to stock/scrap), supplier returns (RMA), warranty claim linkage and status.

- Stock adjustments:
 - Cycle counts, damage/expiry write-offs, transfers between locations, approvals.

Integrations

- Job management: part reservation/consumption against work orders.
- Invoicing: push consumed parts with correct pricing, taxes, discounts.
- Purchasing: PO creation from reorder suggestions or manual demand.
- Accounting (optional): COGS, inventory accounts via configured valuation.
- Vehicle records: fitment and past usage by VIN.

User Roles & Permissions

- Parts Clerk: receive, issue, search, label, count.
- Technician: request/issue to jobs, barcode scan.
- Parts Manager: approve POs/adjustments, edit catalog, set reorder rules.
- Admin: configure valuation, locations, integrations.
- Permissions by action and location; view/edit rights and approvals.

Technical & Data Model

- Entities: Part, Supplier, PriceList, Location, Bin, StockLot/Serial, ReorderRule, PurchaseOrder, GoodsReceipt, WorkOrderIssue, Return (Customer/Supplier), WarrantyClaim, StockAdjustment, Transfer.
- Key fields: Part(SKU, name, category, fitment, unit, barcode, tax, active); StockLot(qty, cost, expiry); ReorderRule(min, max, leadTime, supplier); WorkOrderIssue(part, qty, jobId, price).
- APIs/events for create/update/issue/receive; webhooks for low stock and PO approvals.
- Concurrency control on stock movements; atomic transactions.
- Barcode/QR support; mobile-friendly UI.
- Multi-currency/tax handling; timezone-aware timestamps.

Reporting & Analytics

- Stock on hand by location and value.
- Reorder and backorder lists.
- Usage by period/job/vehicle.
- Slow/fast movers, dead stock.

- Valuation reports (FIFO/WAvg), COGS.
- Discrepancy and audit logs.
- Supplier performance (lead time, fill rate).
- Export to CSV; schedule emails.